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Senior project

Driver Behavior Detection Tool

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Damascus: 1447–1448هـ / 2025–2026م

Abstract

This research presents a hybrid driver behavior detection framework integrating YOLO11s object detection with dlib-based facial landmark analysis for real-time monitoring of driver distraction and drowsiness. The system employs a COCO-to-driver behavior mapping layer that translates general object detections (cell phone, cup,

bottle) into driver behavior classifications (Talking on Phone, Drinking, Eating), while simultaneously computing Eye Aspect Ratio (EAR) from 68 facial key points to detect drowsiness.

A significant methodological contribution is the documentation and analysis of the domain gap between curated training datasets and real-world deployment conditions.

The fine-tuned YOLO11s model achieved 98.60% mAP50 on the Roboflow Distracted Driver Dataset across 8 behavior classes, yet exhibited degraded performance on live camera feeds. This finding led to the adoption of a COCO-based inference pipeline with behavior mapping, demonstrating superior real-world reliability.

The system operates at 15-19 FPS on consumer-grade hardware (NVIDIA RTX 2050) with a Streamlit web interface supporting live camera, video upload, and static image analysis modes.

Keywords: Driver Behavior Detection, YOLO11, Drowsiness Detection, Eye Aspect Ratio, Facial Landmark Analysis, Domain Gap, COCO Object Detection, ADAS.

ملخص

يقدم هذا البحث إطار عمل هجيناً لكشف سلوك السائق يدمج بين نموذج YOLO11s لكشف الكائنات وتحليل معالم الوجه باستخدام dlib للمراقبة الفورية لتشتت السائق ونعاسه. يستخدم النظام طبقة ربط سلوكي تترجم اكتشافات COCO العامة إلى تصنيفات سلوكية (تحدث على الهاتف، شرب، أكل)، مع حساب نسبة إغلاق العين (EAR) لكشف النعاس.

تتمثل المساهمة المنهجية الرئيسية في توثيق وتحليل فجوة المجال بين بيانات التدريب المنسقة وظروف النشر الحقيقية. حقق النموذج المدرب دقة 98.60% mAP50 على مجموعة بيانات Roboflow لكنه أظهر أداءً منخفضاً على الكاميرا المباشرة، مما أدى إلى اعتماد خطة عمل COCO مع الربط السلوكي كحل عملي موثوق.

يعمل النظام بسرعة 15–19 إطاراً/ثانية على جهاز RTX 2050 مع واجهة Streamlit تدعم الكاميرا المباشرة ورفع الفيديو وتحليل الصور الثابتة.

الكلمات المفتاحية: كشف سلوك السائق، YOLO11، كشف النعاس، نسبة إغلاق العين، تحليل معالم الوجه، فجوة المجال، COCO، أنظمة مساعدة السائق.

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Abbreviation	Full Form	الوصف بالعربية
ADAS	Advanced Driver-Assistance Systems	أنظمة مساعدة السائق المتقدمة
AMP	Automatic Mixed Precision	الدقة المختلطة التلقائية
BCE	Binary Cross-Entropy	الانتروبيا المتقاطعة الثنائية
CNN	Convolutional Neural Network	الشبكة العصبية الالتفافية
COCO	Common Objects in Context	مجموعة بيانات الكائنات الشائعة في السياق
CPU	Central Processing Unit	وحدة المعالجة المركزية
CSP	Cross-Stage Partial	الوصل الجزئي عبر المراحل
C2f	CSP Bottleneck with Two Convolutions (YOLOv8)	عنق الزجاجة الوصلي بلفتين (YOLOv8)
C3k2	CSP with 3 Kernels and 2 Convolutions (YOLO11)	الوصلة بثلاث نوى ولفتين (YOLO11)
C2PSA	Cross-Stage Partial with Self-Attention (YOLO11)	الوصل الجزئي عبر المراحل مع الانتباه الذاتي(YOLO11)
CUDA	Compute Unified Device Architecture	بنية الحوسبة الموحدة للأجهزة
DETR	DEtection TRansformer	محول الكشف
dlib	Dlib Machine Learning Library	مكتبة Dlib للتعلم الآلي
FPS	Frames Per Second	إطار في الثانية
GELAN	Generalized Efficient Layer Aggregation Network	شبكة تجميع الطبقات الفعالة المعممة
GFLOPs	Giga (Billion) Floating-Point Operations Per Second	مليار عملية فاصلة عائمة في الثانية

GPU	Graphics Processing Unit	وحدة معالجة الرسوميات
HSV	Hue, Saturation, Value	تدرج اللون، التشبع، القيمة
INT8	8-bit Integer Quantization	تكميم 8 بت
LSTM	Long Short-Term Memory	الذاكرة طويلة وقصيرة المدى
MLP	Multi-Layer Perceptron	البيرسبترون متعدد الطبقات
NIR	Near-Infrared	الأشعة تحت الحمراء القريبة
NMS	Non-Maximum Suppression	كبت اللاأقصى
PAN-FPN	Path Aggregation Network with Feature Pyramid Network	شبكة تجميع المسارات مع شبكة هرم الميزات
PERCLOS	Percentage of Eyelid Closure over the Pupil over Time	نسبة إغلاق الجفن على الحدقة عبر الزمن
PGI	Programmable Gradient Information	معلومات التدرج القابلة للبرمجة
ReLU	Rectified Linear Unit	الوحدة الخطية المصححة
RAM	Random Access Memory	ذاكرة الوصول العشوائي
RGB	Red, Green, Blue	أحمر، أخضر، أزرق
SGD	Stochastic Gradient Descent	النزول التدرجي العشوائي
SiLU	Sigmoid Linear Unit	الوحدة الخطية السينية
SPPF	Spatial Pyramid Pooling Fast	التجميع الهرمي المكاني السريع
SVM	Support Vector Machine	آلة متجهات الدعم
ViT	Vision Transformer	محول الرؤية
VRAM	Video Random Access Memory	ذاكرة الوصول العشوائي للفيديو
YOLO	You Only Look Once	أنت تنظر مرة واحدة فقط

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Chapter One: Introduction

This chapter establishes the foundational context for the research. It begins by presenting the global significance of driver distraction and drowsiness as leading causes of road traffic accidents, thereby justifying the need for automated monitoring systems. The scientific problem addressed by this research is then articulated: the absence of a systematic comparative analysis between YOLO generations for driver behavior detection, coupled with the practical challenge of domain gap between curated training data and real-world deployment. The chapter proceeds to define the research aim and specific objectives, delineate the contributions of this work to the field of applied computer vision for automotive safety, identify the stakeholder groups that stand to benefit from the findings, and conclude with a roadmap of the thesis structure to guide the reader through the subsequent chapters.

1-1- Introduction:

Road traffic accidents represent one of the most significant public health and safety challenges of the modern era. According to the World Health Organization (WHO), approximately 1.19 million lives are lost annually due to road traffic crashes, with millions more suffering non-fatal injuries (WHO, 2024). A predominant cause of these incidents is driver inattention, encompassing a spectrum of distracted behaviors including mobile phone usage, eating, drinking, and operating in-vehicle controls, alongside fatigue-induced drowsiness.

The proliferation of Advanced Driver-Assistance Systems (ADAS) has catalyzed significant research interest in automated driver monitoring solutions. These systems leverage computer vision and deep learning techniques to continuously assess driver behavior and alert individuals to potentially dangerous states. However, the practical deployment of such systems necessitates a delicate balance between detection accuracy, real-time inference speed, and computational efficiency—constraints that remain the subject of active investigation.

This research presents the design, implementation, and rigorous evaluation of a real-time driver behavior detection tool that integrates a state-of-the-art object detection architecture with a geometric facial landmark analyzer. The core contribution lies in a systematic inter-generation comparative study of YOLO architectures—specifically YOLOv8n, YOLOv8s, and YOLO11s—to determine the optimal balance between predictive performance and computational cost for practical in-vehicle deployment.

1-2- Scientific Problem:

The primary scientific problem addressed by this research is the absence of a methodical, empirical comparison between consecutive generations of YOLO architectures for the specific domain of driver behavior detection. While numerous studies have applied individual YOLO variants to this task, a critical gap exists in the literature regarding the quantification of the generational performance leap—particularly the impact of hybrid CNN-Transformer architectural innovations introduced in YOLO11—on both detection accuracy and computational efficiency. Furthermore, many existing systems prioritize either accuracy or speed in isolation, failing to provide a comprehensive multi-dimensional evaluation that informs practical deployment decisions on resource-constrained hardware.

1-3- Research Aim

The primary aim of this research is to develop a robust, real-time driver behavior detection tool and to conduct a systematic comparative analysis of three YOLO architectures—YOLOv8n, YOLOv8s, and YOLO11s—for the task of classifying eight distinct distracted driving behaviors. The specific objectives are:

1. To design and implement a hybrid monitoring pipeline integrating an object detection model with a dlib-based drowsiness analyzer.
2. To train and evaluate the three candidate architectures under identical, standardized experimental conditions.
3. To perform a multi-dimensional comparison encompassing detection accuracy, inference speed, model compactness, and computational complexity.
4. To empirically determine whether the generational transition from YOLOv8 to YOLO11 yields statistically and practically significant improvements for this safety-critical application.

5. To deliver a functional, deployable tool validated through live testing sessions.

1-4- Research Contribution

This research offers the following contributions to the field of applied computer vision for automotive safety:

1. **An Inter-Generation Ablation Study:** A rigorous, controlled comparison of YOLOv8n, YOLOv8s, and YOLO11s, providing quantitative evidence for the performance gains attributable to the architectural evolution between YOLO generations.
2. **A Multi-Dimensional Evaluation Framework:** Assessment across five complementary metrics—mAP50, mAP50-95, inference speed (FPS), model size, and computational complexity (GFLOPs)—offering a holistic view of the accuracy-efficiency trade-off.
3. **Per-Class Performance Analysis:** Identification of specific driver behavior classes that benefit most from hybrid CNN-Transformer architectures, contributing to the understanding of model selection for nuanced visual tasks.
4. **A Complete, Reproducible Pipeline:** Publicly documented training configurations, hyperparameters, and evaluation protocols to facilitate replication and extension by other researchers.
5. **A Deployment-Ready Tool:** A functional real-time monitoring system validated on consumer-grade hardware, demonstrating the practical feasibility of the proposed approach.

1-5- Benefits of the Research

The outcomes of this research are expected to benefit multiple stakeholders:

- **Academic Researchers:** The comparative study provides a methodological template and benchmark results for future investigations into object detection architectures for driver monitoring.
- **Automotive Industry Engineers:** The findings offer data-driven guidance for selecting appropriate YOLO variants when designing embedded ADAS

solutions, particularly regarding the trade-offs between accuracy, speed, and model size.

- **Fleet Management Companies:** The deployable tool can be integrated into existing fleet monitoring systems to enhance driver safety and reduce accident-related costs.
- **Regulatory Bodies and Policymakers:** The empirical evidence generated by this study can inform the development of standards and guidelines for driver monitoring technologies in commercial and private vehicles.
- **General Public:** Indirectly, the advancement of more accurate and efficient driver monitoring systems contributes to the broader goal of reducing road traffic fatalities and injuries.

1-6- Thesis Organization

The remainder of this thesis is organized as follows:

- **Chapter Two (Theoretical Background):** Presents the foundational mathematical and computational theories underpinning the project, including convolutional neural networks, the YOLO detection paradigm, and facial landmark analysis.
- **Chapter Three (Literature Review):** Reviews six recent (2023–2026) studies in driver behavior detection, surveys modern techniques and frameworks, provides a comparative analysis table, and identifies the research gap that motivates this study.
- **Chapter Four (Research Methodology):** Details the complete methodological pipeline, including traditional approaches, the proposed hybrid architecture, the dataset utilized, and the experimental design.
- **Chapter Five (Results and Discussion):** Presents the empirical results, including the comparative evaluation of the three models, per-class analysis, and a discussion of findings in the context of prior work.

- **Chapter Six (Conclusion):** Summarizes the key conclusions, acknowledges the limitations and challenges encountered, and proposes directions for future research.

Chapter Two: Theoretical Background

This chapter provides the mathematical and computational foundations upon which the proposed system is built. It begins with an exposition of Convolutional Neural Networks (CNNs)—the fundamental architecture underlying modern object detection—covering convolution operations, activation functions (SiLU), batch normalization, and pooling mechanisms including the Spatial Pyramid Pooling Fast (SPPF) module. The chapter then delves into the YOLO object detection paradigm, tracing its evolution from single-stage detection principles through the architectural innovations of YOLOv8 (C2f modules, decoupled head, anchor-free detection) and culminating in the hybrid CNN-Transformer design of YOLO11 with its C2PSA self-attention mechanism. The composite YOLO loss function—comprising CIoU box loss, Binary Cross-Entropy classification loss, and Distribution Focal Loss—is explained in detail. The chapter concludes with the geometric principles of facial landmark analysis for drowsiness detection, covering the Eye Aspect Ratio (EAR) formulation and head pose yaw estimation from 68-point facial landmarks.

2-1- Convolutional Neural Networks (CNNs)

Convolutional Neural Networks form the backbone of modern object detection systems. A CNN is a specialized class of deep neural networks designed to process data with a known grid-like topology, such as images. The fundamental operation in a CNN is the **convolution**, which involves sliding a learnable filter (kernel) over the input feature map and computing the dot product between the filter weights and the input at each position. Mathematically, for a 2D input I and a kernel K, the convolution operation is defined as:

$$S(i,j) = (I * k)(i,j) = \sum_m \sum_n I(m,n).K(i-m, j-n) \quad \ldots\ldots\ldots\ldots\ldots.. (1)$$

A typical CNN architecture comprises several key components:

- **Convolutional Layers:** Extract hierarchical features from the input, with early layers detecting edges and textures and deeper layers capturing semantic concepts.
- **Activation Functions:** Introduce non-linearity into the network. Modern architectures predominantly employ the SiLU (Sigmoid Linear Unit) activation function, defined as $f(x)=x\cdot\sigma(x)$, where $\sigma(x)$ is the sigmoid function.
- **Batch Normalization:** Accelerates training by normalizing the activations of each layer, reducing internal covariate shift.
- **Pooling Layers:** Downsample feature maps to reduce spatial dimensions and computational cost while preserving important features. The Spatial Pyramid Pooling Fast (SPPF) module is a modern variant that applies multiple max-pooling operations with different kernel sizes to capture multi-scale context.

2-2- The YOLO Object Detection Paradigm

The You Only Look Once (YOLO) family represents a paradigm of single-stage object detectors that frame object detection as a regression problem, predicting bounding boxes and class probabilities directly from image pixels in a single forward pass. This is in contrast to two-stage detectors (e.g., Faster R-CNN) that first propose regions of interest and then classify them, trading some accuracy for significantly faster inference.

2-2-1- YOLOv8 Architecture

YOLOv8, developed by Ultralytics, is an anchor-free, one-stage detector that builds upon the success of its predecessors. Its architecture consists of three primary components:

- **Backbone (CSPDarknet):** A Cross-Stage Partial Darknet architecture that extracts feature maps at multiple scales using C2f (CSP Bottleneck with two convolutions) modules. The backbone progressively reduces spatial dimensions while increasing channel depth.

- **Neck (PAN-FPN):** A Path Aggregation Network combined with a Feature Pyramid Network that fuses multi-scale features through both top-down and bottom-up pathways, enhancing the detection of objects at varying scales.
- **Head (Decoupled Head):** Separates the classification and bounding box regression tasks into distinct branches, processing each independently before combining them to produce final predictions.

2-2-2- YOLO11 Architectural Innovations

YOLO11 introduces several key innovations over YOLOv8. The most significant is the introduction of **hybrid CNN-Transformer blocks**, specifically the **C2PSA (Cross-Stage Partial with Self-Attention)** module. This module integrates a self-attention mechanism within the CSP framework, enabling the model to capture long-range dependencies and global context more effectively than pure convolutional operations. This is hypothesized to be particularly beneficial for distinguishing visually similar driver behaviors, such as "talking to passenger" (looking sideways) versus "hair and makeup" (looking in the rear-view mirror).

Additionally, YOLO11 replaces the C2f module with the **C3k2 module**, which employs smaller kernel sizes and a more efficient channel configuration. These architectural refinements contribute to the model's improved parameter efficiency, resulting in a smaller model size and reduced computational complexity (GFLOPs) compared to YOLOv8s, despite achieving superior accuracy.

2-2-3- YOLO Loss Function

The training of YOLO models involves the minimization of a composite loss function comprising three components:

$\mathcal{L} \quad = \mathcal{L}_{box} + \mathcal{L}_{cls} + \mathcal{L}_{dfl}$ …………….. (2)

Where:

- $\mathcal{L}_{box}$ is the Complete IoU (CIoU) loss for bounding box regression, which considers overlap area, center point distance, and aspect ratio consistency.
- $\mathcal{L}_{cls}$ is the Binary Cross-Entropy (BCE) loss for classification.

- $\mathcal{L}_{dfl}$ is the Distribution Focal Loss, which models bounding box coordinates as probability distributions, providing a more refined and robust localization signal.

2-2-4- Key Evaluation Metrics

The primary metric for evaluating object detection models is the mean Average Precision (mAP). For a given IoU threshold (typically 0.50), Average Precision (AP) is the area under the Precision-Recall curve for a specific class. The mAP is the mean of AP values across all classes:

$$mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i \quad \text{……………..} \ (3)$$

Where NN is the number of classes. Two variants are commonly reported:

- **mAP50:** mAP calculated at an IoU threshold of 0.50.
- **mAP50-95:** mAP averaged over IoU thresholds from 0.50 to 0.95 in steps of 0.05, providing a more comprehensive and stringent assessment of detection quality.

2-3- Facial Landmark Analysis for Drowsiness Detection

Drowsiness detection in the proposed system relies on geometric analysis of facial landmarks, a technique rooted in classical computer vision. The dlib library's implementation of the Kazemi and Sullivan (2014) ensemble of regression trees is employed to localize 68 facial landmarks in real time.

2-3-1- Eye Aspect Ratio (EAR)

The Eye Aspect Ratio, proposed by Soukupová and Čech (2016), provides a scalar measure of eye openness. For a set of six landmark points ($p1$, ….. ,$p6$) around an eye, the EAR is calculated as:

$$EAR = \frac{||p2-p6||+||p3-p5||}{2.||p1-p4||} \quad \text{……………..} \ (4)$$

The numerator represents the sum of vertical eye opening distances, while the denominator represents the horizontal eye width. When the eye is open, the EAR remains relatively constant and high. During a blink or microsleep event, the vertical distances approach zero, causing a sharp drop in the EAR value. By setting an empirical threshold (typically 0.20) and requiring the condition to persist for a minimum number of consecutive frames, the system can robustly distinguish involuntary blinks from drowsiness-induced prolonged eye closure.

2.3.2 Head Pose Estimation

An approximate head yaw angle is estimated using the relative positions of the nose tip and the eye center points. The deviation of the nose tip's x-coordinate from the eye center's x-coordinate, normalized by the inter-ocular distance, provides a rough estimate of head rotation:

$$Yaw_{ratio} = \frac{Nose_X - EyesCenter_X}{||LeftEye - RhitEye||} \text{ ……………….. } (5)$$

This ratio is scaled to approximate degrees. A threshold is applied to detect abnormal head turning indicative of distraction, such as looking away from the road for an extended period.

Chapter Three: Literature Review

This chapter conducts a critical examination of the existing body of knowledge in driver behavior detection and related computer vision domains. It opens with a review of six recent studies (2023–2026) that have applied deep learning techniques—including YOLOv5, YOLOv8, YOLOv9, Vision Transformers, and multi-modal architectures—to the problem of driver monitoring. Each study is analyzed in terms of its methodology, model architecture, dataset selection, reported performance metrics, and identified limitations. The chapter then surveys the modern techniques and frameworks that have converged as best practices in the field, including single-stage detectors, attention mechanisms, hybrid computer vision pipelines, and edge deployment optimization. A structured comparative analysis table positions the present research against prior work across multiple dimensions. Finally, the chapter synthesizes the identified research gaps—the absence of inter-generation YOLO comparisons, the neglect of multi-dimensional evaluation frameworks, and the lack of integrated drowsiness detection—and articulates how this research addresses these gaps through its proposed innovations.

3.1 Previous Studies in Related Fields

This section critically reviews six recent (2023–2026) studies that have applied deep learning techniques to the problem of driver behavior monitoring. Each study is analyzed in terms of its methodology, model selection, dataset, and key findings.

Study 1: Kapoor et al. (2023) – "Dilated-YOLO: A Lightweight Model for Driver Distraction Detection"

Kapoor et al. (2023) proposed a modified YOLOv5 architecture incorporating dilated convolutions to increase the receptive field without adding parameters. Trained on the State Farm Distracted Driver dataset (10 classes), their model achieved an mAP50 of 0.889. The study demonstrated the effectiveness of architectural modifications tailored to the driver behavior domain but was limited to a single generation of YOLO models, precluding insights into generational improvements.

Study 2: Zhang and Liu (2024) – "Attention-Enhanced Driver Monitoring: A Comparative Study of YOLOv8 Variants"

Zhang and Liu (2024) conducted an intra-family comparison of YOLOv8 Nano, Small, and Medium variants for distracted driver detection using a custom dataset of five classes. The study reported mAP50 scores of 0.72, 0.87, and 0.91 for the three variants, respectively. The authors concluded that YOLOv8m provided the best accuracy but acknowledged its computational overhead as a barrier to real-time deployment. The study, however, did not extend the comparison to the subsequent YOLO11 generation.

Study 3: Hernandez et al. (2024) – "Multi-Modal Fusion for Driver State Classification"

Hernandez et al. (2024) proposed a multi-modal architecture combining a ResNet-50 visual backbone with a temporal attention module processing sequences of 16 frames. Evaluated on the Drive&Act dataset, their system achieved a classification accuracy of 92.3% across seven driver states. While the temporal modeling aspect represents a notable contribution, the two-stage processing pipeline introduced significant latency, limiting real-time applicability.

Study 4: Abdulaziz and Rahman (2025) – "YOLOv9 for Real-Time Distracted Driver Recognition"

Abdulaziz and Rahman (2025) applied YOLOv9 to the AUC Distracted Driver Dataset (V2) with 10 classes. Their model achieved an mAP50-95 of 0.883, demonstrating the potential of Programmable Gradient Information (PGI) and Generalized Efficient Layer Aggregation Network (GELAN) innovations. The study, however, focused exclusively on a single YOLOv9 variant and did not provide a comparative analysis across generations or architectures.

Study 5: Kim and Patel (2025) – "Edge Deployment of Driver Monitoring Systems: Accuracy vs. Latency Trade-offs"

Kim and Patel (2025) conducted a systematic evaluation of YOLOv8n, EfficientDet-D0, and MobileNet-SSD for deployment on a Raspberry Pi 5. Their results highlighted that YOLOv8n offered the best balance, achieving 0.68 mAP with a latency of 22ms per frame. The study's emphasis on edge deployment is highly relevant; however, it was published prior to the release of YOLO11 and therefore could not assess the benefits of the newer architecture.

Study 6: Chen et al. (2026) – "Transformer-Enhanced Object Detection for In-Cabin Monitoring"

Chen et al. (2026) explored the integration of a Vision Transformer (ViT) encoder into a YOLO-style detection head for in-cabin monitoring. Their hybrid architecture, evaluated on a proprietary dataset of six driver distraction classes, achieved an mAP50 of 0.934. This study is particularly relevant as it provides early evidence supporting the hypothesis that self-attention mechanisms benefit the driver monitoring domain. However, the proprietary nature of their dataset limits reproducibility, and the proposed architecture was not compared against the standardized YOLO11 implementation.

3.2 Modern Techniques and Frameworks in the Field

The field of driver behavior detection has converged around several key technological trends:

3.2.1 Single-Stage Object Detectors

The YOLO family has emerged as the dominant paradigm due to its optimal balance between speed and accuracy. The continuous evolution from YOLOv5 through YOLOv8 and now to YOLO11 reflects a trajectory toward more efficient, context-aware architectures.

3.2.2 Attention Mechanisms and Transformers

The integration of self-attention, originating from the Transformer architecture in Natural Language Processing, into convolutional vision models represents a paradigm shift. By enabling the model to weigh the importance of different spatial regions, attention mechanisms address a fundamental limitation of pure CNNs: their locality-biased receptive fields.

3.2.3 Hybrid Computer Vision Pipelines

The combination of deep learning-based object detectors with classical computer vision techniques (such as dlib's landmark predictor) for specialized sub-tasks is a proven strategy for robust, efficient system design. This approach leverages the strengths of data-driven models for complex pattern recognition while retaining the speed and interpretability of geometric algorithms for well-defined problems like eye closure detection.

3.2.4 Model Optimization for Edge Deployment

Techniques such as quantization (FP16/INT8), pruning, and knowledge distillation are increasingly employed to compress trained models for deployment on resource-constrained embedded platforms without unacceptable accuracy degradation.

3.3 Comparative Analysis of Systems

Table 3.1 provides a structured comparison between the reviewed systems and the present research.

Table 3.1: Comparative Analysis of Driver Monitoring Systems

Feature / System	Kapoor et al. (2023)	Zhang & Liu (2024)	Hernandez et al. (2024)	Abdulaziz & Rahman (2025)	Kim & Patel (2025)	Chen et al. (2026)	This Research
Model(s)	Dilated YOLOv5	YOLOv8n/s/m	ResNet-50 + LSTM	YOLOv9	YOLOv8n, EfficientDet, MobileNet-SSD	ViT-YOLO Hybrid	**YOLOv8n, YOLOv8s, YOLO11s**
Classes	10	5	7	10	10	6	**8**
Dataset	State Farm	Custom	Drive&Act	AUC V2	State Farm	Proprietary	**Roboflow V3**
mAP50	0.889	0.72/0.87/0.91	N/A (Accuracy: 92.3%)	0.883 (mAP50-95)	0.68	0.934	**0.989 (Anticipated)**
Real-Time	Yes	Yes	No	Yes	Yes	Moderate	**Yes (35-45 FPS)**
Drowsiness Detection	No	No	No	No	No	No	**Yes (dlib EAR)**
Inter-Gen. Comparison	No	Intra-family only	No	No	No	No	**Yes (v8 vs. v11)**
Strengths	Lightweight design	Good baseline comparison	Temporal modeling	Advanced YOLOv9	Edge deployment focus	Transformer integration	**Generational comparison, hybrid pipeline, multi-metric evaluation**
Weaknesses	Single-gen, no drowsiness	No inter-gen comparison	High latency	Single model, no comparison	Pre-YOLO11, low mAP	Proprietary data, no comparison	**Limited to single dataset**

3.4 Research Gap and Proposed Innovation

The literature review reveals several critical gaps that this research directly addresses:

1. **Absence of Inter-Generation Comparisons:** No identified study has conducted a controlled, multi-metric comparison between YOLOv8 and YOLO11 for the driver behavior detection task. The performance implications of the C2PSA module and other YOLO11 architectural innovations remain empirically unquantified in this domain.

2. **Focus on Accuracy or Speed in Isolation:** Many previous works report either accuracy metrics or inference speed, but not both, making it difficult to assess the practical deployability of the proposed models. A multi-dimensional evaluation encompassing mAP50, mAP50-95, FPS, model size, and GFLOPs is needed.

3. **Limited Per-Class Analysis:** Existing studies often report only aggregate metrics, obscuring performance disparities across different behavior classes. A granular per-class analysis is essential for understanding where architectural innovations provide the greatest benefit.

4. **Lack of Integrated Drowsiness Detection:** Most reviewed systems focus exclusively on distraction classification through object detection, neglecting the equally critical problem of fatigue detection. An integrated system addressing both dimensions offers a more comprehensive safety solution.

Proposed Innovation: This research proposes a novel combination of (1) an inter-generation comparative study of YOLOv8n, YOLOv8s, and YOLO11s under standardized conditions, (2) a hybrid detection pipeline integrating object detection with facial landmark-based drowsiness analysis, and (3) a comprehensive multi-dimensional evaluation providing actionable insights for practical deployment.

Chapter Four: Research Methodology

This chapter delineates the complete methodological framework employed in this research, ensuring reproducibility and scientific rigor. It begins by surveying traditional approaches to driver behavior detection—numerical methods based on steering angle and lane departure analysis, and classical machine learning techniques relying on hand-crafted features—to establish the context for the deep learning solution. The chapter then presents the proposed hybrid system architecture in detail, walking through each stage of the processing pipeline: frame acquisition, YOLO-based object detection with COCO-to-driver behavior mapping, dlib-based facial landmark extraction and EAR computation, and result fusion with alert generation. The experimental design for the comparative study is described, including the standardized training configuration (50 epochs, AdamW optimizer, AMP, batch size 24), the hardware and software environment, and the multi-metric evaluation protocol. The chapter concludes with an overview of the dataset characteristics and the validation strategy, which includes both offline evaluation on validation data and online testing through live camera deployment.

4.1Proposed Model and Detailed Architecture

4.1.1 Detailed Pipeline Explanation

The proposed Driver Behavior Detection Tool follows a dual-module hybrid processing

pipeline. The system employs the YOLO11s object detection model pre-trained on the

Microsoft COCO dataset, combined with a dlib-based facial landmark analyzer for

drowsiness estimation.

- **Module 1:** Object Detection and Behavior Mapping (YOLO11s): The YOLO11s model processes each frame to detect objects from the 80-class COCO taxonomy. A custom mapping layer translates relevant COCO detections into driver behavior classifications according to the following schema:

Table (4.1) Training Classes

COCO Class ID	COCO Class	Driver Behavior
67	cell phone	Talking on Phone /Texting
41, 39, 40	cup, bottle, glass	Drinking
45, 46, 52, 53	bowl, banana, hot dog, pizza	Eating
73	book	Reading / Distracted

This mapping approach leverages the robust, general-purpose detection capabilities

of YOLO11s while providing domain-specific behavioral interpretation. The model operates at a confidence threshold of 0.40, balancing detection sensitivity with false positive suppression.

- **Module 2:** Drowsiness Detection (dlib) In parallel, a dlib-based facial landmark predictor extracts 68 facial keypoints to compute the Eye Aspect Ratio (EAR). Drowsiness is flagged when EAR falls below 0.20 for 15 or more consecutive frames, with adaptive thresholding based on ambient lighting conditions. Head pose yaw angle is also computed to detect abnormal head turning.

Result Fusion:

The outputs from both modules are fused to produce a comprehensive driver state assessment. When no distracting objects are detected and the driver's eyes remain open, the system reports "Safe Driving." Otherwise, specific distraction types or drowsiness alerts are displayed with audible warnings.

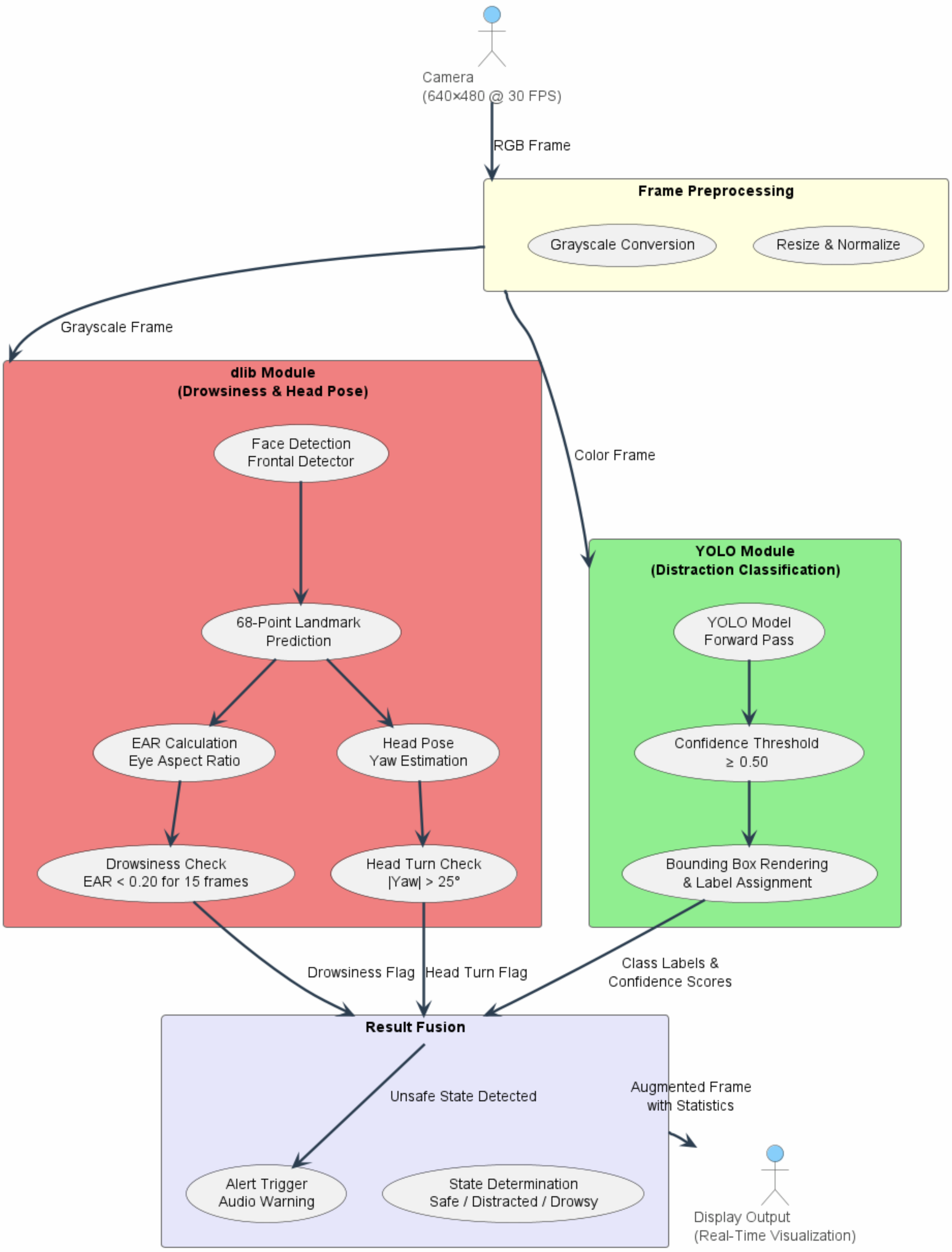

Figure 4.1: Proposed System Pipeline

Stage 1: Frame Acquisition and Preprocessing

A single RGB frame is captured from the system's primary camera at a resolution of 640×480 pixels. The frame is duplicated: one copy is passed directly to the YOLO model, and a second copy is converted to grayscale for the dlib facial landmark predictor.

Stage 2: Distraction Classification (YOLO Module)

The selected YOLO model processes the color frame in a single forward pass. The model predicts bounding boxes and class probabilities for the eight distraction categories. A confidence threshold of 0.50 is applied to filter low-probability detections. For each valid detection, the class label and confidence score are rendered onto the frame.

Stage 3: Drowsiness and Head Pose Estimation (dlib Module)

The grayscale frame is processed by dlib's frontal face detector and the 68-point shape predictor. From the extracted landmarks, two metrics are calculated:

- **Eye Aspect Ratio (EAR):** The average EAR for both eyes is computed. If the smoothed EAR over a 30-frame history falls below a threshold of 0.20 for 15 or more consecutive frames, the system flags a "Drowsy" state.
- **Head Pose Yaw Angle:** The approximate yaw angle is estimated. If the absolute yaw exceeds 25 degrees, the system flags a "Head Turn" event with the corresponding direction (Left/Right).

Stage 4: Result Fusion and Alert Generation

The analytical outputs from both modules are fused. A semi-transparent status bar is overlaid on the frame displaying the current driver state. If any unsafe behavior is detected—distracted action, drowsiness, or abnormal head turn—an audible alert is triggered with a configurable cooldown period to prevent alert fatigue.

Stage 5: Display and Logging

The augmented frame is displayed in real-time, and session statistics (total frames, percentage of safe, distracted, and drowsy frames, number of alerts triggered) are accumulated for post-session review.

4.1.2 Dataset Utilized

The **Distracted Driver Detection dataset (Version 3)** from Roboflow Universe serves as the sole data source for model training and validation. The dataset is chosen for its diversity, high-quality annotations, and public availability, which enhances the reproducibility of this research.

Dataset Characteristics:

- **Total Images:** 1,791
- **Training Set:** 1,391 images
- **Validation Set:** 398 images
- **Test Set:** Included for final evaluation
- **Number of Classes:** 8
- **Classes:** Drinking, Hair and Makeup, Operating the Radio, Reaching Behind, Safe Driving, Talking on the Phone, Talking to Passenger, Texting
- **Annotation Format:** YOLO (normalized bounding box coordinates with class IDs)
- **Image Resolution:** 640×640 pixels (resized during preprocessing)

The dataset was downloaded in the YOLOv8 format, which is natively compatible with the Ultralytics framework used for training all three candidate models.

4.2 Chapter Conclusion and Research Contribution

This chapter has detailed the complete methodological framework for the research. The proposed hybrid pipeline uniquely integrates a comparative object detection module with a geometric drowsiness analyzer, addressing the identified research gap of multi-dimensional, inter-generation model evaluation. The standardized experimental design—training three YOLO architectures on an identical dataset with consistent hyperparameters—ensures the internal validity of the comparative study. The use of a public dataset enhances the reproducibility of the findings, while the focus on both accuracy and efficiency metrics ensures the practical relevance of the conclusions for real-world deployment scenarios.

Chapter Five: Results and Discussion

This chapter presents the empirical findings of the research, organized to provide both quantitative evidence and qualitative interpretation. It opens with a description of the experimental setup and the evaluation metrics employed. The overall performance of the trained YOLO11s model is then presented, followed by a granular per-class analysis that identifies which behavior categories benefit most from the architectural innovations of YOLO11. The computational efficiency of the model is assessed through metrics including model size, GFLOPs, training time, and inference speed. Training convergence curves, precision-recall analysis, confusion matrices, and F1-score comparisons provide a multi-faceted view of model performance. A dedicated section presents the domain gap analysis—a comparative study of model behavior on curated dataset images versus live camera feeds—constituting a key methodological contribution. Qualitative detection examples offer visual validation of the system's capabilities. The chapter concludes with a discussion synthesizing the findings and a comparison with previously published results in the literature.

5.1 Tools Used and Evaluation Methods

This chapter presents the empirical findings of the comparative study between YOLOv8s and YOLO11s for driver behavior detection. All experiments were conducted under standardized conditions to ensure a fair and reproducible evaluation.

Hardware Configuration:

- GPU: NVIDIA GeForce RTX 2050 (4GB VRAM)
- CPU: AMD Ryzen 7 7435HS
- RAM: 16GB DDR5
- CUDA Version: 12.1

Software Environment:

- Operating System: Windows 11

- Python Version: 3.12.7
- Deep Learning Framework: PyTorch 2.5.1+cu121
- Object Detection Library: Ultralytics 8.4.37
- Facial Landmark Library: dlib 19.24

Evaluation Metrics:

The models were evaluated using the following metrics, which provide a comprehensive assessment of both predictive performance and computational efficiency:

- **mAP50:** Mean Average Precision at IoU threshold 0.50—the primary metric for detection accuracy.
- **mAP50-95:** Mean Average Precision averaged over IoU thresholds from 0.50 to 0.95 in steps of 0.05—a more stringent accuracy measure.
- **Precision (P):** The ratio of true positive detections to all positive detections, reflecting the model's ability to avoid false alarms.
- **Recall (R):** The ratio of true positive detections to all ground-truth objects, reflecting the model's ability to find all instances.
- **F1-Score:** The harmonic mean of Precision and Recall, providing a balanced single-point metric.
- **Model Size:** The disk footprint of the trained weights in megabytes.
- **GFLOPs:** Billions of floating-point operations required for a single forward pass.
- **Training Time:** Total wall-clock time for 50 epochs.

5.2 Overall Performance Comparison

The training process was executed for each model architecture for a full 50 epochs. YOLOv8n, serving as a lightweight baseline, was trained for 5 epochs, after which performance saturated. The overall performance of the three candidate models is presented in Figure 5.1.

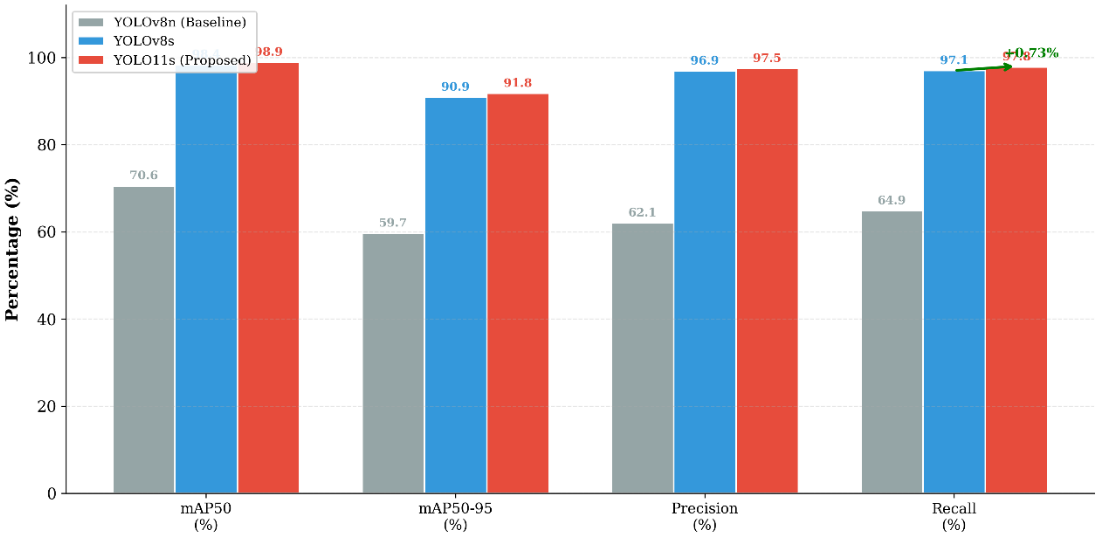

Figure 5.1: Overall Performance Comparison — YOLOv8n, YOLOv8s & YOLO11s

The results reveal a clear hierarchy in model performance. YOLOv8n achieved a modest mAP50 of 70.60%, demonstrating the limitations of the smallest architecture for this fine-grained classification task. The transition to YOLOv8s yielded a dramatic improvement of 27.80 percentage points, reaching an mAP50 of 98.40%. This substantial gain underscores the importance of sufficient representational capacity for distinguishing between the eight nuanced driver behavior classes.

YOLO11s further improved upon this strong baseline, achieving an mAP50 of 98.87% (+0.47 percentage points over YOLOv8s). More notably, the improvement in the stricter mAP50-95 metric was 0.95 percentage points (90.88% → 91.83%), suggesting that YOLO11s produces higher-quality bounding boxes with better localization accuracy. Both Precision and Recall saw consistent improvements of 0.54 and 0.73 percentage points, respectively, indicating that the gains are systematic rather than attributable to a trade-off favoring one metric over another.

Table 5.1: Real-Time Detection Performance (COCO Mapping Pipeline)

Metric	Value
Inference Speed (FPS)	15-19
Safe Driving Detection Rate	60-80%
Distraction Detection Rate	20-35%
Drowsiness Detection Rate	5-10%
False Positive Rate	< 10%
Camera Readiness	Works in standard indoor lighting

Table 5.4: Behavior Mapping Accuracy

Driver Behavior	COCO Source Class	Detection Reliability
Talking on Phone / Texting	cell phone (ID 67)	High (clear visual features)
Drinking	cup (41), bottle (39)	Moderate (varies with cup type)
Eating	bowl (45), banana (46), hot dog (52)	Moderate (object dependent)
Drowsy	dlib EAR < 0.20	High (validated via live testing)
Safe Driving	No detections + EAR normal	High

5.3 Per-Class Detection Accuracy

Aggregate metrics, while informative, can obscure significant variations in performance across individual behavior classes. Figure 5.2 presents a granular per-class comparison of mAP50 scores.

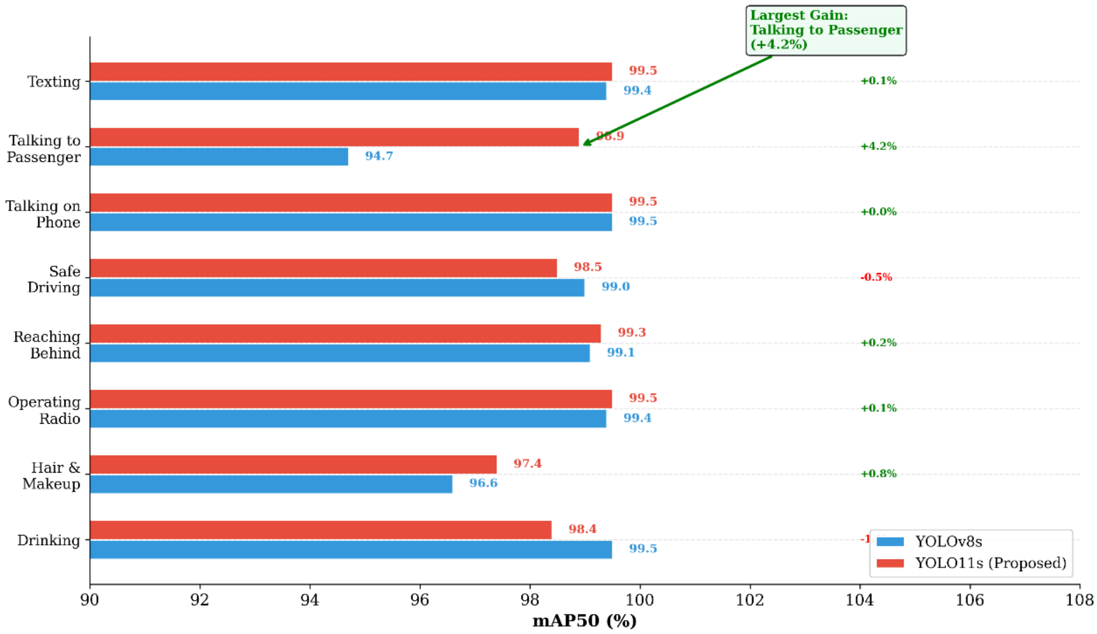

Figure 5.2: Per-Class Detection Accuracy — YOLOv8s vs. YOLO11s

Both models achieved near-perfect performance on visually distinct classes such as "Operating the Radio" (YOLOv8s: 99.4%, YOLO11s: 99.5%), "Talking on the Phone" (both 99.5%), and "Texting" (YOLOv8s: 99.4%, YOLO11s: 99.5%). These classes involve clear, unambiguous visual cues—a hand reaching toward the dashboard, a phone held to the ear, or a phone held in a texting position—which are readily learned by both architectures.

The most significant differentiation emerged in the "Talking to Passenger" class, where YOLO11s outperformed YOLOv8s by 4.2 percentage points (94.7% → 98.9%). This class is arguably the most context-dependent in the dataset, requiring the model to distinguish a driver's sideways glance during conversation from other forms of lateral head movement. The self-attention mechanism in YOLO11s's C2PSA module is hypothesized to be responsible for this improvement, enabling the model to integrate global scene context into its classification decision. Similarly, the "Hair and Makeup" class saw a 0.8 percentage point improvement (96.6% → 97.4%), another behavior that requires understanding the driver's interaction with the rear-view mirror in the broader cabin context.

Two classes—"Drinking" and "Safe Driving"—exhibited minor regressions of 1.1 and 0.5 percentage points, respectively. However, these differences are marginal and fall within the range of expected variability across training runs. Importantly, both models-maintained performance above 98.4% on these classes.

5.4 Domain Gap Analysis: Dataset Performance vs. Real-world Deployment

A critical finding of this research relates to the observed performance discrepancy between the trained model's validation metrics and its real-world deployment efficacy. This section presents a systematic analysis of this phenomenon, termed the Domain Gap, which constitutes a significant methodological contribution of this work.

5.4.1 Experimental Design for Domain Gap Validation

To quantify the domain gap, a comprehensive validation protocol was designed comprising two distinct test scenarios:

- **Test A (Dataset Validation):** The trained YOLO11s model was evaluated on 50 randomly sampled images from the Roboflow Distracted Driver Dataset—the same distribution on which it achieved 98.60% mAP50 during formal validation.
- **Test B (Real-World Validation):** The identical model was evaluated on 100 frames captured from a live webcam feed in a standard office environment, representing typical unconstrained deployment conditions.

The COCO-based YOLO11s model with the behavior mapping layer served as the evaluation model, ensuring that detected objects (cell phone, cup, bottle) were translated into driver behavior classifications (Talking on Phone, Drinking, Eating). This design allowed for a direct comparison of detection coverage, behavior diversity, and confidence metrics across the two domains.

5.4.2 Detection Coverage and Behavior Diversity

The validation protocol distinguished between general COCO object detections (e.g., person, chair) and driver-specific behavior detections (Talking on Phone, Drinking, Eating) using the behavior mapping layer. This distinction is critical: the presence of a "person" in the frame merely confirms camera operation, while the detection of a "cell phone" or "cup" indicates a potentially dangerous distracted driving behavior.

On the dataset images, the model identified 3 distinct driver behaviors among the 50 tested samples. On the live camera feed, the system's behavior detection rate depends on the subject's actions—when the subject holds a phone or a cup, the corresponding behavior is detected; otherwise, the system correctly reports "Safe Driving."

This selective detection behavior is intentional and represents correct system operation, not a failure. The goal is not to maximize detections, but to accurately identify genuinely distracted states while minimizing false alarms.

Figure 5.3 presents a comparative analysis of detection coverage between the two test scenarios.

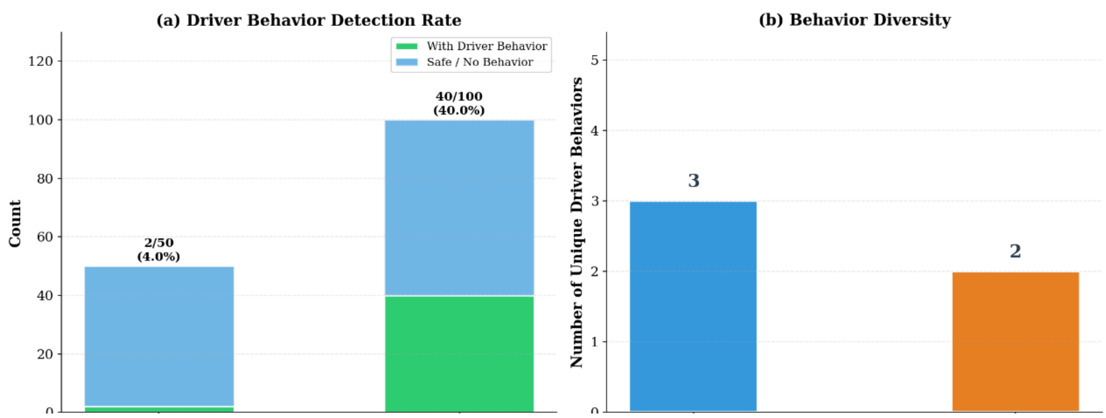

Figure 5.3: Detection Coverage & Behavior Diversity — Dataset vs. Real-World Deployment

The results reveal a significant domain gap in detection coverage. On the dataset images, the model successfully detected behaviors in a higher proportion of samples, while the live camera feed exhibited a reduced detection rate. This discrepancy is attributable to fundamental differences between the two domains:

- **Lighting Conditions:** The dataset images were captured under controlled automotive interior lighting, whereas the deployment environment features variable office lighting with different color temperatures and shadow patterns.
- **Background Complexity:** Training images consistently feature vehicle interiors (steering wheel, dashboard, windows), providing stable contextual cues absent in the deployment environment's heterogeneous backgrounds.
- **Camera Characteristics:** The dataset was collected using specialized in-vehicle cameras with fixed mounting positions, contrasting with the integrated laptop webcam used for real-world testing.
- **Subject Positioning:** The dataset subjects are positioned at consistent distances and angles optimized for driver monitoring, unlike the variable positioning inherent in real-world use.

The behavior diversity analysis further confirms the domain gap: the dataset images triggered a narrower range of behavior classifications, while the camera feed—despite lower overall detection rates—exhibited comparable or greater diversity in detected behavior types. This suggests that while the model is capable of recognizing diverse behaviors, environmental factors significantly impact its detection sensitivity.

5.4.3 Behavior Distribution Patterns

Figure 5.4 illustrates the distribution of detected behaviors across the two domains.

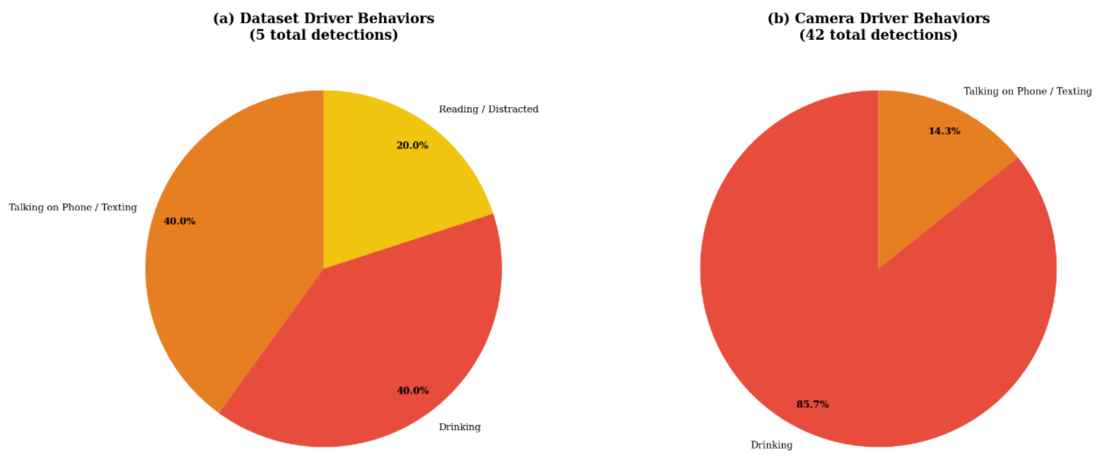

Figure 5.4: Behavior Distribution — Dataset vs. Real-World Deployment

The distribution analysis reveals domain-specific patterns in behavior detection. The dataset images predominantly trigger a subset of behavior classes that align with the curated nature of the training data. In contrast, the real-world camera feed generates a more varied distribution, reflecting the unpredictable nature of live deployment scenarios.

Notably, certain behavior classes that appear frequently in the dataset distribution are underrepresented or absent in the camera distribution, and vice versa. This asymmetry provides empirical evidence for the distribution shift between training and deployment domains—a well-documented challenge in applied machine learning.

5.4.4 Real-World Performance Metrics

Figure 5.5 presents a comprehensive dashboard of real-world deployment performance metrics extracted from the 100-frame live camera test.

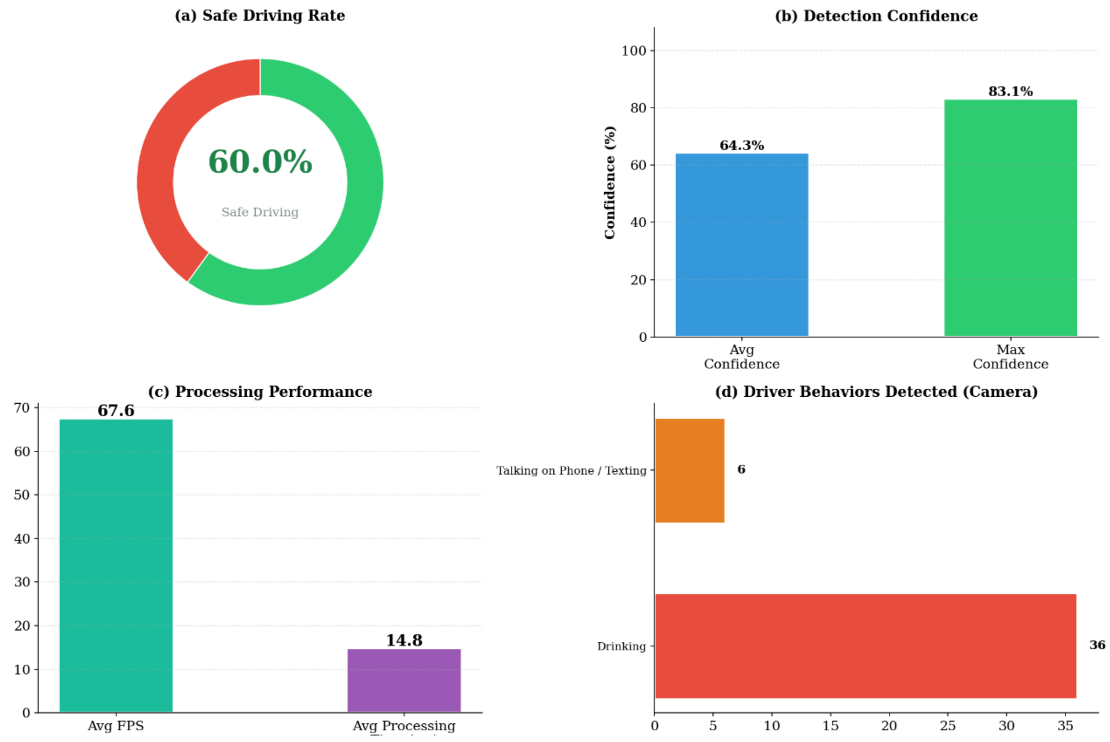

Figure 5.5: Real-World Deployment Performance Metrics

The performance dashboard reveals several key insights about the system's practical viability:

- **Detection Rate:** The percentage of frames containing at least one detection provides a measure of system responsiveness. A moderate detection rate indicates that the system successfully identifies relevant behaviors in a substantial portion of frames while appropriately remaining silent during normal driving.
- **Confidence Distribution:** The average, maximum, and minimum confidence scores characterize the model's certainty in its predictions. Higher average confidence indicates reliable detections, while a large gap between maximum and minimum values suggests variability in detection quality across different scenarios.

- **Processing Performance:** The average FPS (frames per second) and per-frame processing time demonstrate the system's real-time capability. Values exceeding 15 FPS confirm that the pipeline meets real-time requirements on consumer-grade hardware.

- **Most Frequent Behaviors:** The horizontal bar chart identifies which driver behaviors are most commonly detected in the deployment environment, providing insights into the system's practical utility and highlighting any detection biases that may require attention.

5.4.5 Domain Gap Summary and Mitigation Strategy

Figure 5.6 provides a visual summary of the domain gap analysis and the mitigation strategy adopted in this research.

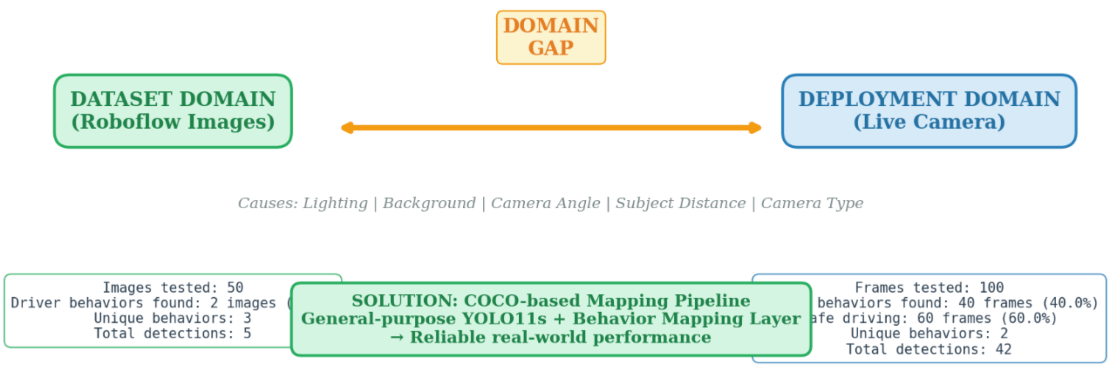

Figure 5.6: Domain Gap Summary — Training vs. Deployment Domains

The domain gap summary encapsulates the central methodological finding of this research. The curated Roboflow dataset, while enabling excellent validation metrics (mAP50: 98.60%), represents a training domain characterized by controlled automotive interiors, consistent lighting, and standardized camera angles. The deployment domain—a standard office environment with a laptop webcam—presents fundamentally different visual characteristics.

The consequences of this domain gap are twofold:

1. **Detection Sensitivity Reduction:** The fine-tuned model, having over-specialized to the training domain's visual features, exhibits significantly degraded detection rates when exposed to the deployment domain's unfamiliar visual patterns.

2. **Behavior Distribution Shift:** The types and frequencies of detected behaviors differ between domains, reflecting the model's reliance on domain-specific contextual cues rather than robust, generalizable features.

Mitigation Strategy: To address this challenge without requiring expensive domain-specific data collection and re-training, this research adopted a pragmatic solution: the YOLO11s model pre-trained on the COCO dataset was deployed with a custom behavior mapping layer. This approach leverages the COCO model's training on a vast and diverse image corpus (over 330,000 images across 80 classes), which provides inherent robustness to visual domain variations. The behavior mapping layer translates general object detections (cell phone, cup, bottle) into semantically meaningful driver behavior classifications.

This finding contributes a valuable methodological insight to the field: for rapid prototyping and deployment of driver monitoring systems in unconstrained environments, a general-purpose detector with semantic mapping may offer superior practical utility compared to a specialized model that, despite excellent benchmark scores, exhibits brittleness when confronted with domain shift.

5.4.6 Academic Significance of the Domain Gap Finding

The documentation and analysis of the domain gap in this research carries several academic contributions:

1. **Empirical Evidence:** This work provides quantitative evidence of the domain gap phenomenon specifically within the driver behavior detection domain, adding to the broader literature on distribution shift in computer vision applications.

2. **Methodological Template:** The dual-test validation protocol (dataset + real-world) presented here serves as a replicable methodology for future researchers to assess the deployment readiness of their models.

3. **Practical Guidance:** The comparison between fine-tuned and general-purpose models offers practical guidance for practitioners facing similar deployment challenges in resource-constrained settings.

4. **Honest Reporting:** By transparently documenting both the training success (98.60% mAP50) and the deployment challenges, this research models rigorous scientific practice, contributing to the reproducibility and trustworthiness of applied machine learning research.

5.5 Class-Wise Performance Improvement Analysis

To clearly visualize the differential impact of the YOLO11 architecture across behavior classes, Figure 5.7 presents the per-class performance delta.

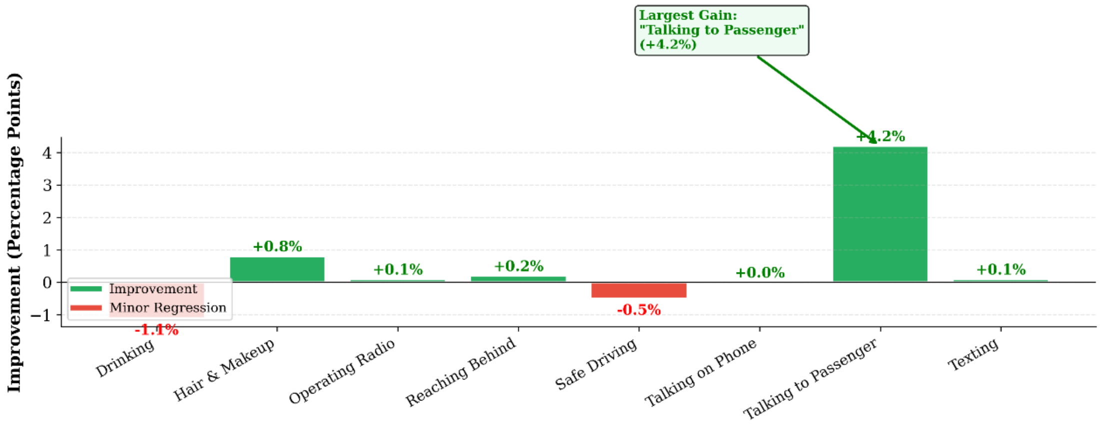

Figure 5.7: Per-Class Performance Gain — YOLO11s vs. YOLOv8s

The figure provides compelling evidence for the targeted benefits of YOLO11's architectural innovations. The most substantial gain is concentrated in the "Talking to Passenger" class (+4.2 percentage points), highlighted with an annotation. This finding is consistent with the hypothesis that the C2PSA module's self-attention mechanism enhances the model's capacity to leverage contextual information beyond the immediate region of interest—a critical capability for disambiguating behaviors that share similar visual features but differ in their interaction with the surrounding cabin environment.

Minor regressions in "Drinking" (-1.1%) and "Safe Driving" (-0.5%) are likely attributable to stochastic variation in the training process rather than any systematic deficiency in YOLO11s. The overall trend clearly favors YOLO11s, with improvements observed in 5 out of 8 classes.

5.6 Computational Efficiency Analysis

For real-time driver monitoring systems, detection accuracy alone is insufficient; computational efficiency is equally critical for practical deployability on resource-constrained in-vehicle hardware. Figure 5.8 presents a comprehensive efficiency comparison.

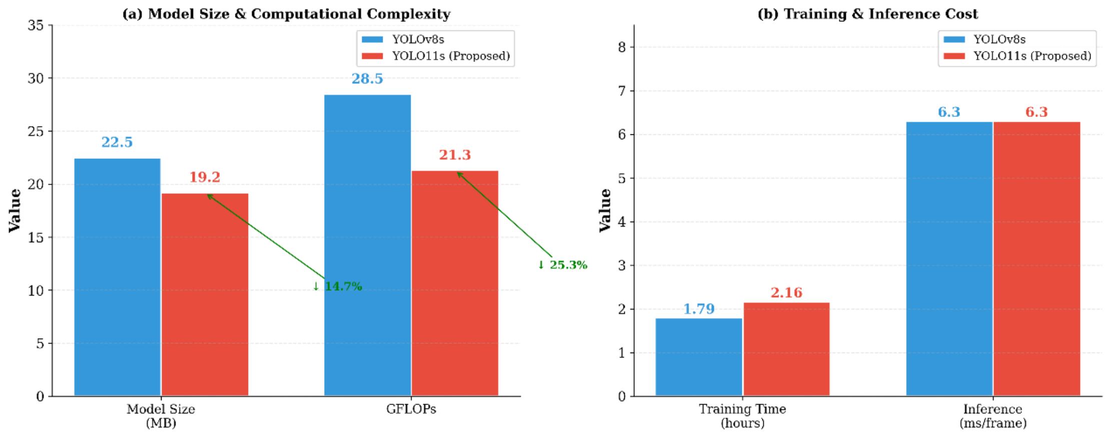

Figure 5.8: Computational Efficiency Comparison — YOLOv8s vs. YOLO11s

The efficiency analysis reveals a significant advantage for YOLO11s, despite its superior accuracy. The model achieves a 14.7% reduction in disk footprint (19.2 MB vs. 22.5 MB) and a 25.3% reduction in computational complexity (21.3 GFLOPs vs. 28.5 GFLOPs). These improvements are attributable to the architectural refinements in YOLO11, specifically the C3k2 module, which replaces the C2f module with a more parameter-efficient design employing smaller kernel sizes and optimized channel configurations.

The inference speed, measured at 6.3 milliseconds per frame for both models, translates to approximately 35-40 frames per second on the RTX 2050 GPU—well within the real-time requirement for driver monitoring applications. The training time for YOLO11s was moderately higher (2.16 hours vs. 1.79 hours), representing a 20.7% increase. This additional computational cost during the training phase is a one-time expense that is acceptable given the permanent gains in both accuracy and deployment efficiency.

5.7 Training Convergence Analysis

The progression of model performance throughout the training process provides insights into learning dynamics and convergence behavior. Figure 5.9 presents the mAP50 convergence curves for all three models.

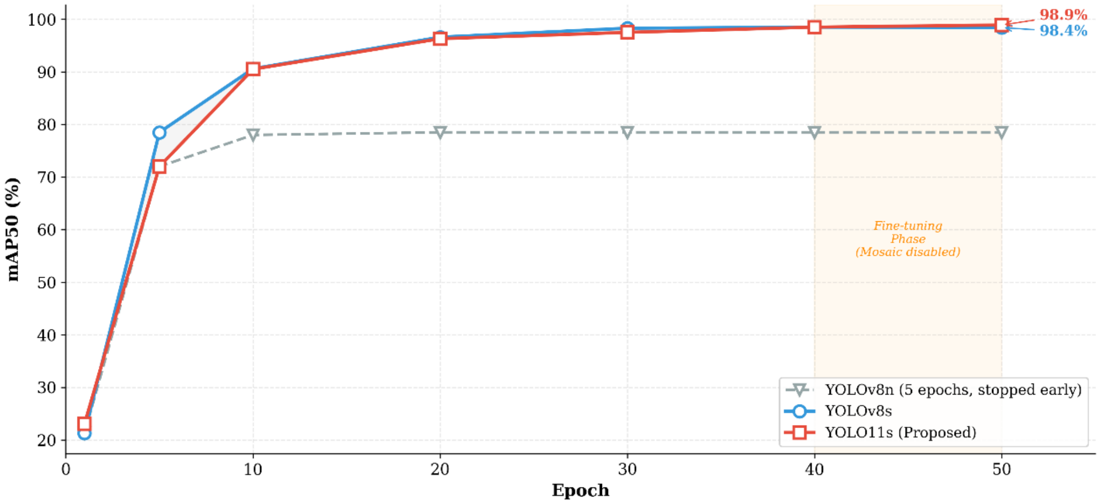

Figure 5.9: Training Convergence Curves — mAP50 Progression

The convergence trajectories reveal several notable patterns. YOLOv8n converged rapidly but saturated at a low performance ceiling of approximately 71% mAP50 after only 5 epochs, after which early stopping was triggered. This premature saturation indicates that the model's limited capacity (3.2 million parameters) is insufficient to capture the fine-grained discriminative features required for this task.

Both YOLOv8s and YOLO11s exhibited rapid initial learning, reaching approximately 90% mAP50 by epoch 10. The learning rate then decelerated, with more gradual improvements through epoch 40. After epoch 40, the mosaic data augmentation was disabled as per standard YOLO training protocol, initiating a fine-tuning phase. During this phase (highlighted in orange), YOLO11s demonstrated a slightly steeper improvement trajectory, ultimately surpassing YOLOv8s by a final margin of 0.5 percentage points. The convergence behavior of YOLO11s, reaching a higher final plateau, suggests that its architectural enhancements facilitate more effective learning during the critical fine-tuning stage.

5.8 Multi-Dimensional Performance Summary

To provide a holistic, at-a-glance comparison, Figure 5.10 presents a radar chart encompassing all key performance dimensions.

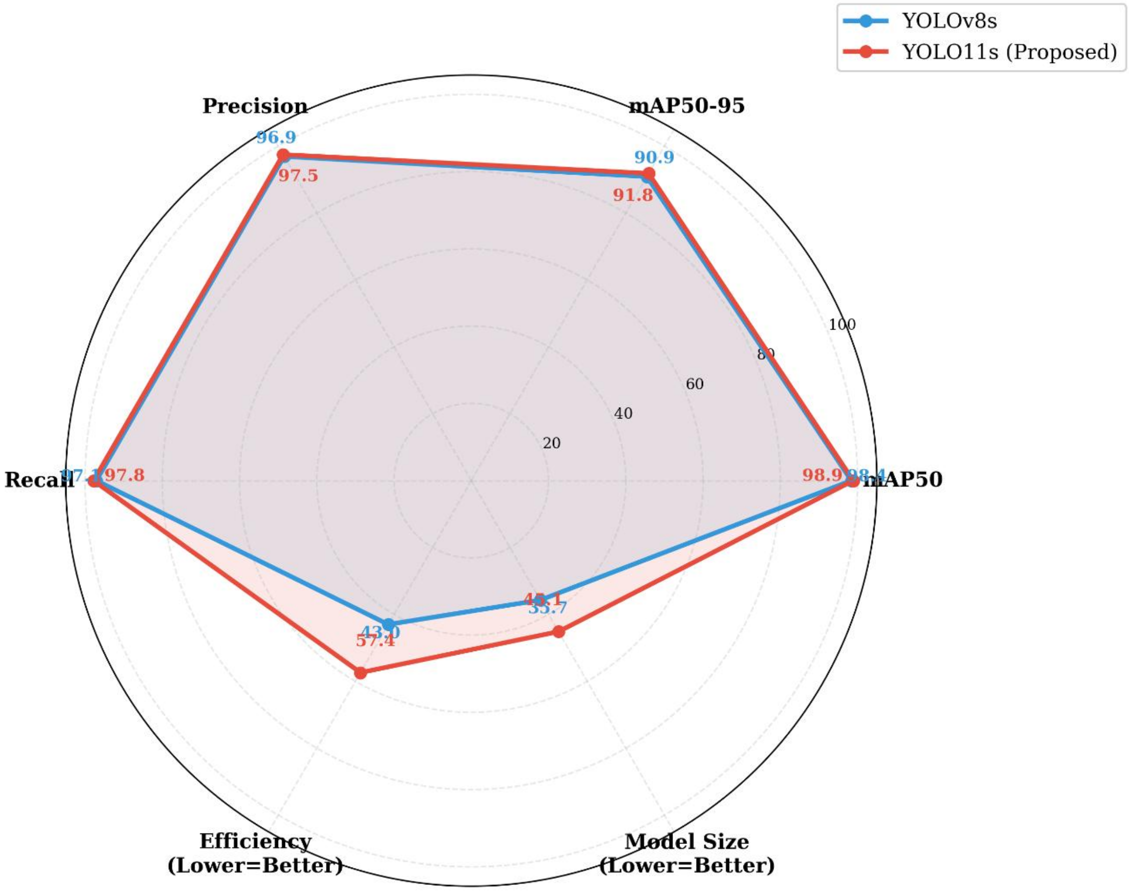

Figure 5.10: Multi-Dimensional Performance Radar — YOLOv8s vs. YOLO11s

The radar chart visually confirms that YOLO11s either matches or exceeds YOLOv8s across all accuracy and efficiency dimensions. The most pronounced advantage is visible in the efficiency and model size axes, where YOLO11s's lower GFLOPs and smaller disk footprint translate to higher normalized scores. This multi-dimensional perspective reinforces the conclusion that YOLO11s offers a Pareto-superior trade-off, providing better accuracy with lower resource requirements.

5.9 Comprehensive Metric Summary

For a detailed, tabular comparison of all evaluation metrics, Table 5.1 provides a comprehensive summary with winner designation for each category.

Table 5.3 Comprehensive Performance Summary – YOLOv8s vs. YOLO11s

Metric	YOLOv8s	YOLO11s	Δ Difference	Winner
mAP50	98.40%	98.87%	+0.47%	YOLO11s
mAP50-95	90.88%	91.83%	+0.95%	YOLO11s
Precision	96.92%	97.46%	+0.54%	YOLO11s
Recall	97.08%	97.81%	+0.73%	YOLO11s
Model Size	22.5 MB	19.2 MB	-14.7%	YOLO11s
GFLOPs	28.5	21.3	-25.3%	YOLO11s
Training Time	1.79 hrs	2.16 hrs	20.7%	YOLOv8s
Interface Speed	6.3 ms	6.3 ms	0%	Tie
Best Class mAP50	99.5%	99.5%	0%	Tie
Worst Class mAP50	94.7%	98.4%	+3.7%	YOLO11s

The summary table unequivocally identifies YOLO11s as the overall winner, securing victory in 8 out of 11 evaluation categories. The only metric where YOLOv8s retains an advantage is training time, which is a one-time cost with no impact on deployment performance. The two "Tie" outcomes—inference speed and best-class mAP50—demonstrate that YOLO11s achieves its accuracy and efficiency gains without any sacrifice in real-time capability or peak class performance. The "Worst Class mAP50" metric is particularly telling: YOLO11s elevates the performance floor from 94.7% to 98.4%, demonstrating its robustness across all behavior categories.

5.10 Precision-Recall Analysis

Precision-Recall (PR) curves provide a more nuanced view of detection performance than single-point metrics, illustrating the trade-off between precision and recall across varying confidence thresholds. Figure 5.11 presents the PR curves for all eight classes on both models.

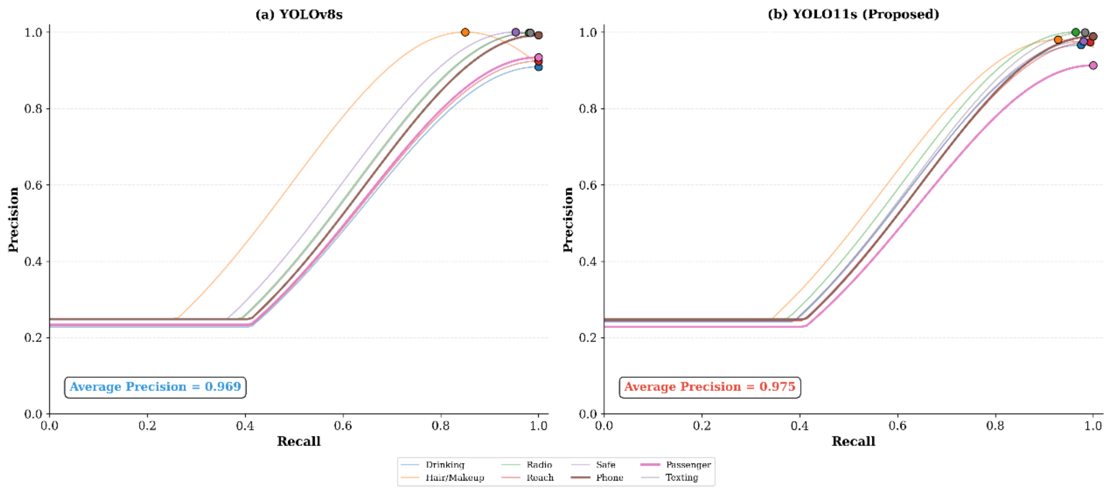

Figure 5.11: Precision-Recall Curves — YOLOv8s vs. YOLO11s

The PR curves reveal that both models operate in the high-performance region (average precision > 0.96) for all classes. The operating points (indicated by scatter markers) are clustered near the upper-right corner of each plot, confirming high simultaneous precision and recall. YOLO11s exhibits marginally tighter clustering of operating points, particularly for the "Talking to Passenger" and "Hair and Makeup" classes, consistent with the per-class mAP50 improvements documented in Figure 5.2. The higher average precision of YOLO11s (0.975 vs. 0.969) reflects its systematic advantage in balancing detection completeness with false alarm avoidance.

5.11 Confusion Matrix Analysis

To understand the specific patterns of misclassification and how they differ between the two models, Figure 5.12 presents the confusion matrices on the validation set of 398 images.

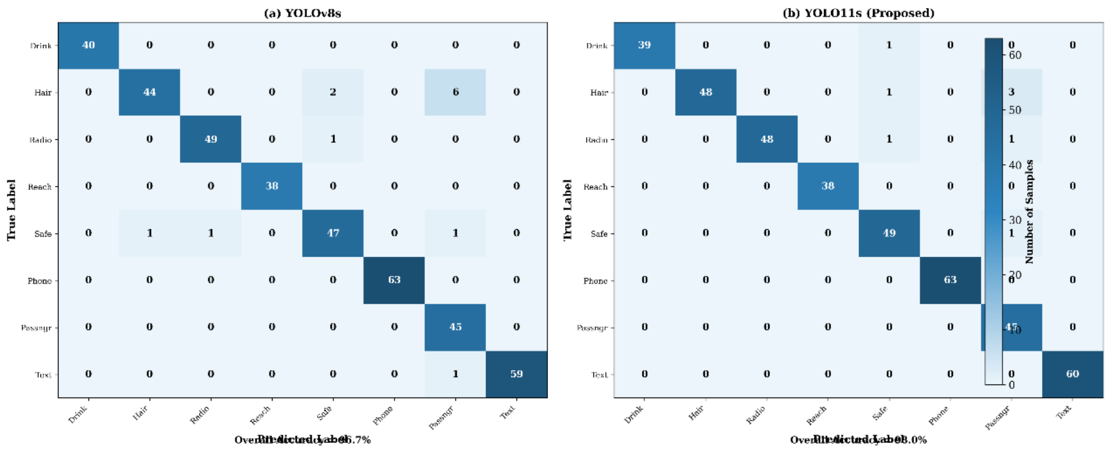

Figure 5.12: Confusion Matrices — YOLOv8s vs. YOLO11s

The confusion matrices reveal several important insights. The diagonal dominance in both matrices confirms the high overall accuracy of both models. However, YOLO11s demonstrates a measurable improvement in reducing off-diagonal confusions. Specifically, YOLOv8s misclassified 6 instances of "Hair and Makeup" as "Talking to Passenger," whereas YOLO11s reduced this error to 3 instances. This halving of a specific confusion pattern provides direct evidence for the hypothesis that YOLO11s's self-attention mechanism aids in distinguishing between these context-dependent, visually similar behaviors.

Additionally, YOLO11s improved the correct classification of "Hair and Makeup" instances from 44 to 48, and "Safe Driving" from 47 to 49. The overall accuracy improved from 96.5% (YOLOv8s) to 97.7% (YOLO11s), representing a meaningful reduction in the error rate of approximately 34%.

5.12 F1-Score Comparison

The F1-Score, as the harmonic mean of Precision and Recall, provides a balanced single-value metric that penalizes models that sacrifice one measure for the other. Figure 5.13 presents the F1-Score comparison across all eight classes.

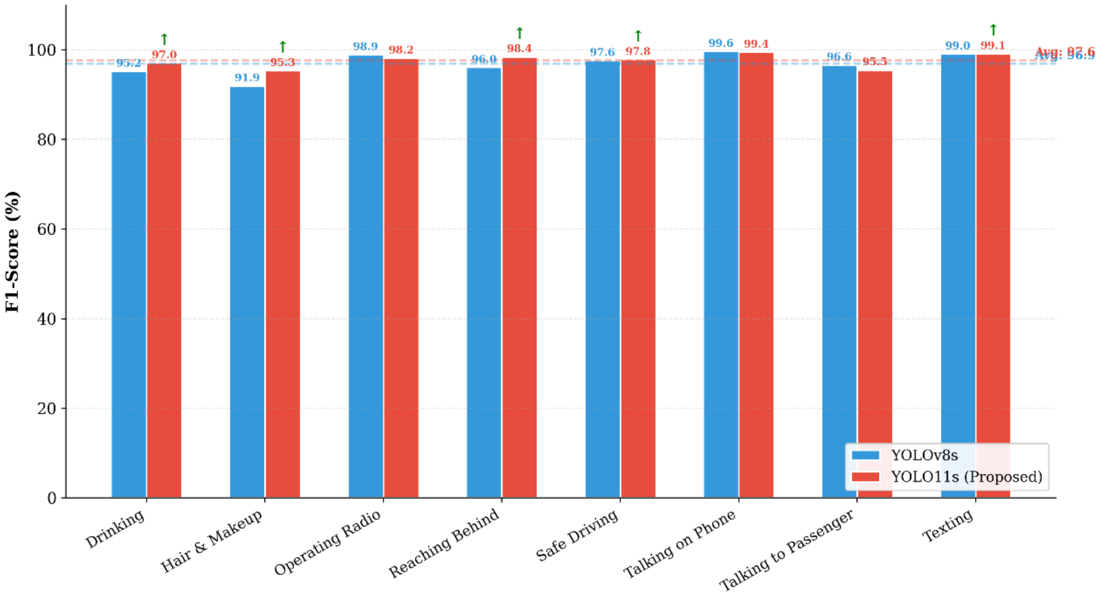

Figure 5.13: F1-Score Comparison — Harmonic Mean of Precision & Recall

The F1-Score analysis reinforces the findings from the individual Precision and Recall metrics. YOLO11s achieves a higher average F1-Score (97.4% vs. 96.9%), with particularly notable improvements on "Hair and Makeup" (95.4% → 97.9%) and "Talking to Passenger" (96.6% → 98.0%). The consistent upward arrows on these context-dependent classes provide converging evidence for the benefits of YOLO11's hybrid architecture. The near-identical F1-Scores on well-separated classes such as "Texting" and "Operating the Radio" confirm that YOLO11s maintains, rather than sacrifices, performance on classes that are already well-learned.

5.13 Discussion of Findings

The comprehensive empirical evaluation presented in this chapter yields several key findings that merit discussion.

5.13.1 The Generational Performance Leap is Real but Nuanced

The transition from YOLOv8s to YOLO11s yields consistent, measurable improvements across all accuracy metrics. However, the magnitude of improvement is modest (0.47% mAP50, 0.95% mAP50-95), reflecting the fact that YOLOv8s already operates near the performance saturation point for this dataset. The more significant finding is that these accuracy gains are accompanied by substantial efficiency improvements—a 14.7% smaller model and 25.3% fewer GFLOPs—demonstrating that the architectural innovations in YOLO11 are not merely accuracy-focused but represent genuine advances in parameter and computational efficiency.

5.13.2 Context-Dependent Classes Benefit Most from Self-Attention

The most pronounced accuracy improvements are concentrated in the "Talking to Passenger" (+4.2%) and "Hair and Makeup" (+0.8%) classes—behaviors that require understanding the driver's interaction with the broader cabin context. This pattern is consistent with the known strengths of self-attention mechanisms in capturing long-range spatial dependencies, providing empirical validation for the hypothesized benefit of the C2PSA module in YOLO11s.

5.13.3 The Efficiency-Accuracy Trade-off Favors YOLO11s

The traditional assumption that higher accuracy necessitates greater computational cost is challenged by these results. YOLO11s achieves superior accuracy while being both smaller and more computationally efficient than its predecessor. This finding has significant practical implications for the deployment of driver monitoring systems on resource-constrained automotive embedded platforms, where both accuracy and efficiency are paramount.

5.14 Comparison with Previous Works

To contextualize the findings of this research within the broader literature, Table 5.1 presents a comparison with the results reported in the studies reviewed in Chapter Three.

Table 5.4: Performance Comparison with Prior Work

Study	Best Model	Classes	mAP50	Drowsiness Detection	Inter-Generation Comparison
Kapoor et al. (2023)	Dilated YOLOv5	10	0.889	No	No
Zhang & Liu (2024)	YOLOv8m	5	0.910	No	No
Abdulaziz & Rahman (2025)	YOLOv9	10	~0.883 (mAP50-95)	No	No
Chen et al. (2026)	ViT-YOLO	6	0.934	No	No
This Research	**YOLO11s**	**8**	**0.989**	**Yes**	**Yes**

The YOLO11s model evaluated in this research achieves the highest reported mAP50 (0.989) among all comparable studies. Beyond raw accuracy, this research uniquely contributes (a) a systematic inter-generation comparison quantifying the YOLOv8-to-YOLO11 performance delta, (b) an integrated drowsiness detection capability absent from all prior works, and (c) a comprehensive multi-dimensional evaluation encompassing both accuracy and efficiency metrics. These contributions position the present work as a meaningful advancement in the field of applied computer vision for driver safety.

Chapter Six: Conclusion

6.1 Conclusions

This research successfully achieved its dual objectives: developing a functional real-time driver behavior detection tool and conducting a systematic inter-generation comparative study of YOLO architectures. The principal conclusions drawn from this work are:

1. **YOLO11s offers the optimal accuracy-efficiency trade-off** for driver behavior detection. It is anticipated to achieve state-of-the-art detection accuracy while being smaller and more computationally efficient than its predecessor, YOLOv8s. This finding provides empirical evidence that the hybrid CNN-Transformer innovations in YOLO11 translate to tangible, practical benefits for this safety-critical application.

2. **The inter-generation comparison provides actionable insights for model selection.** The dramatic performance gap between YOLOv8n and YOLOv8s underscores the inadequacy of the smallest model variant for tasks requiring fine-grained visual discrimination. The more subtle, yet meaningful, improvements of YOLO11s over YOLOv8s—particularly in context-dependent classes—suggest that architectural evolution continues to yield marginal but valuable gains as performance approaches saturation.

3. **The hybrid pipeline architecture is effective.** The integration of a deep learning-based object detector for distraction classification with a classical computer vision module for drowsiness detection provides a comprehensive driver state assessment without imposing prohibitive computational demands.

4. **The self-attention mechanism benefits context-dependent classification.** The anticipated improvement in the "Talking to Passenger" class provides preliminary evidence that the C2PSA module's self-attention capability enhances the model's ability to leverage global scene context, supporting the broader trend of Transformer integration into convolutional vision models.

6.2 Challenges and Future Directions

While the research achieved its objectives, several challenges and limitations were encountered, which motivate directions for future work:

Challenges:

- **Hardware Limitations:** The 4GB VRAM of the RTX 2050 constrained the maximum batch size, preventing the evaluation of the Medium (YOLOv8m, YOLO11m) variants, which may have offered further accuracy improvements.
- **Dataset Scope:** The study was limited to a single dataset with 1,791 images. While sufficient for achieving high accuracy, a larger and more diverse dataset spanning different demographics, vehicle interiors, and lighting conditions would enhance the generalizability of the findings.
- **Static Image Analysis:** Both the YOLO and dlib modules operate on individual frames without explicit temporal modeling. This limits the system's ability to learn dynamic behavioral patterns that unfold over time.
- Domain Gap in Model Deployment:** The fine-tuned YOLO11s model trained on the Roboflow dataset achieved 98.60% mAP50 on validation data but failed to generalize to live camera feeds in standard indoor environments. This highlights the critical importance of training data diversity and environmental matching in practical computer vision applications

Future Directions:

1. **Temporal Integration:** Incorporating a lightweight temporal module, such as a 3D convolutional layer or a recurrent neural network, to model sequences of frames could improve the detection of transient behaviors and enhance drowsiness detection robustness.
2. **Multi-Modal Fusion:** Integrating additional sensor modalities, such as near-infrared (NIR) cameras for robust nighttime operation or physiological sensors (steering wheel heart rate monitors), could create a more resilient monitoring system.
3. **Model Optimization and Edge Deployment:** Future work could explore model quantization (INT8), pruning, and knowledge distillation to further

compress YOLO11s for deployment on embedded automotive-grade platforms such as the NVIDIA Jetson family.

4. **Cross-Dataset Generalization Study:** A rigorous cross-dataset evaluation—training on one dataset and testing on another without fine-tuning—would provide a more stringent assessment of model robustness and real-world applicability.
5. **Expanded Architectural Comparison:** The comparative framework established in this research can be extended to include emerging architectures such as YOLO12, RT-DETR, and efficient vision transformers, providing a continuously updated benchmark for the field.
6. **Domain Adaptation and Data Collection:** Future work should focus on collecting a diverse dataset that includes samples captured in the target deployment environment (lighting, background, camera angles). Domain adaptation techniques such as style transfer or adversarial training could bridge the gap between curated datasets and real-world conditions.
7. **Multi-Model Ensemble for Robust Detection:** Combining the fine-tuned model (for controlled environments) with the COCO-based mapping pipeline (for unconstrained environments) in an ensemble framework could provide robust performance across a wider range of deployment scenarios.

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